



Group Assignment 2 Presentation and Demo



Team 5

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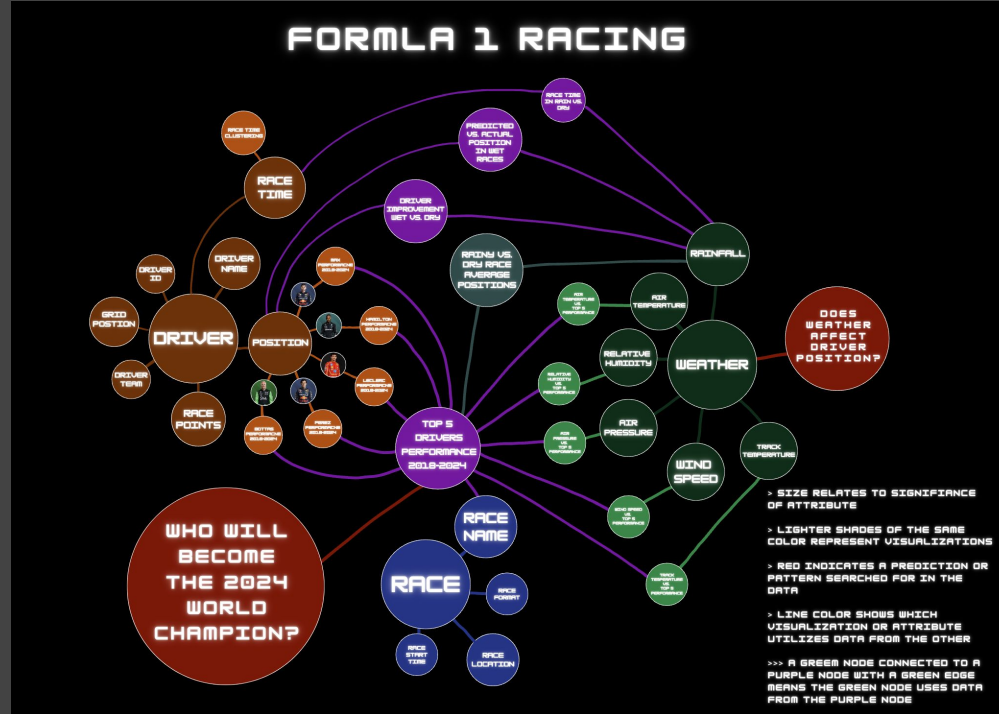
Formula 1 Datasets

Before:

- The dataset only had races from the year 2024 and very limited information about the actual races.
- There was no tracking of what the weather conditions were on specific race dates
- Null data points in the nationality category in previous years of races

Updates:

- There are now weather attributes(rainfall, air pressure, relative humidity, and temperature)
- Removal of diver nationality due to previous years not listing the nationalities
- Adding years as early as 2018 in order to ensure that there is a large sample of data.



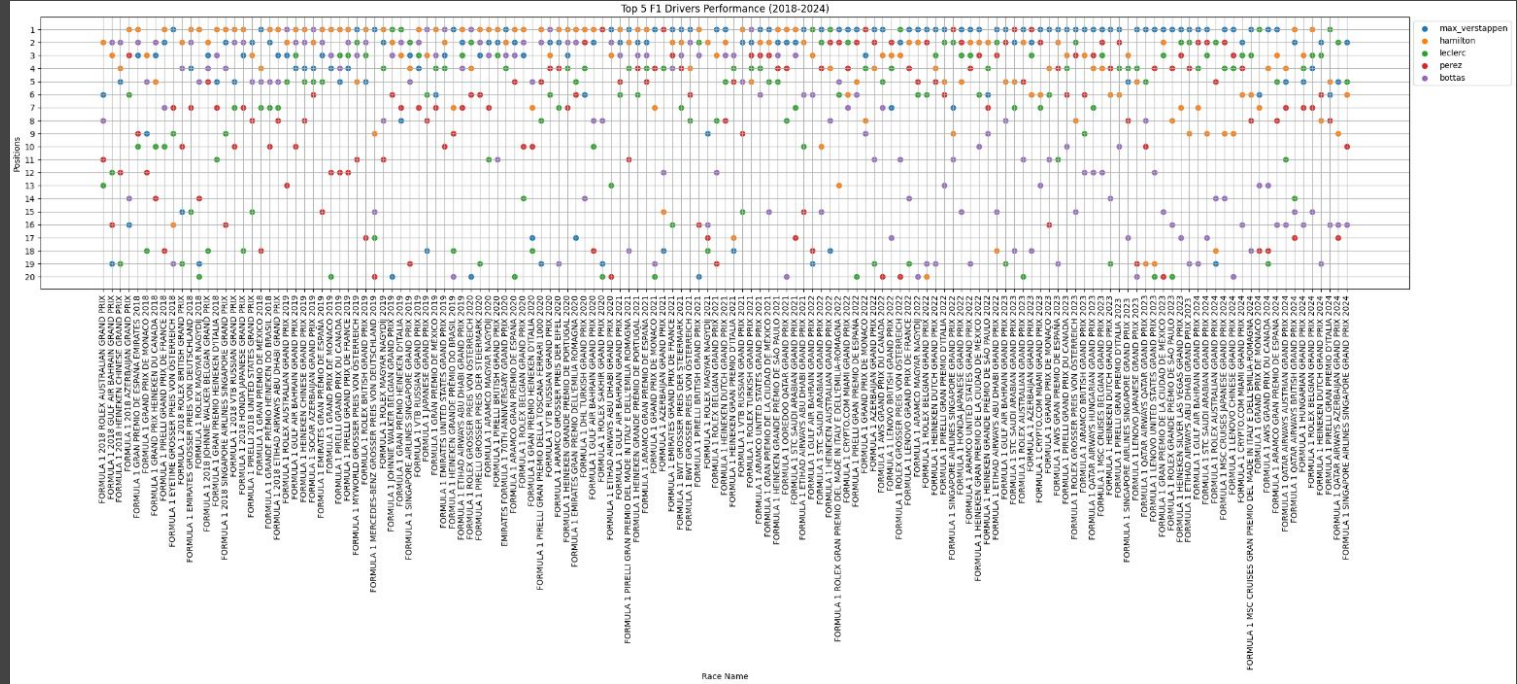
Note : for a more clear, in-depth view of the infographic, please view the submitted pdf or visit this link

https://www.canva.com/design/DAGSohi7c2s/eLPsPwRn7Supe-e-6pNAXA/view?utm_content=DAGSohi7c2s&utm_campaign=designshare&utm_medium=link&utm_source=editor

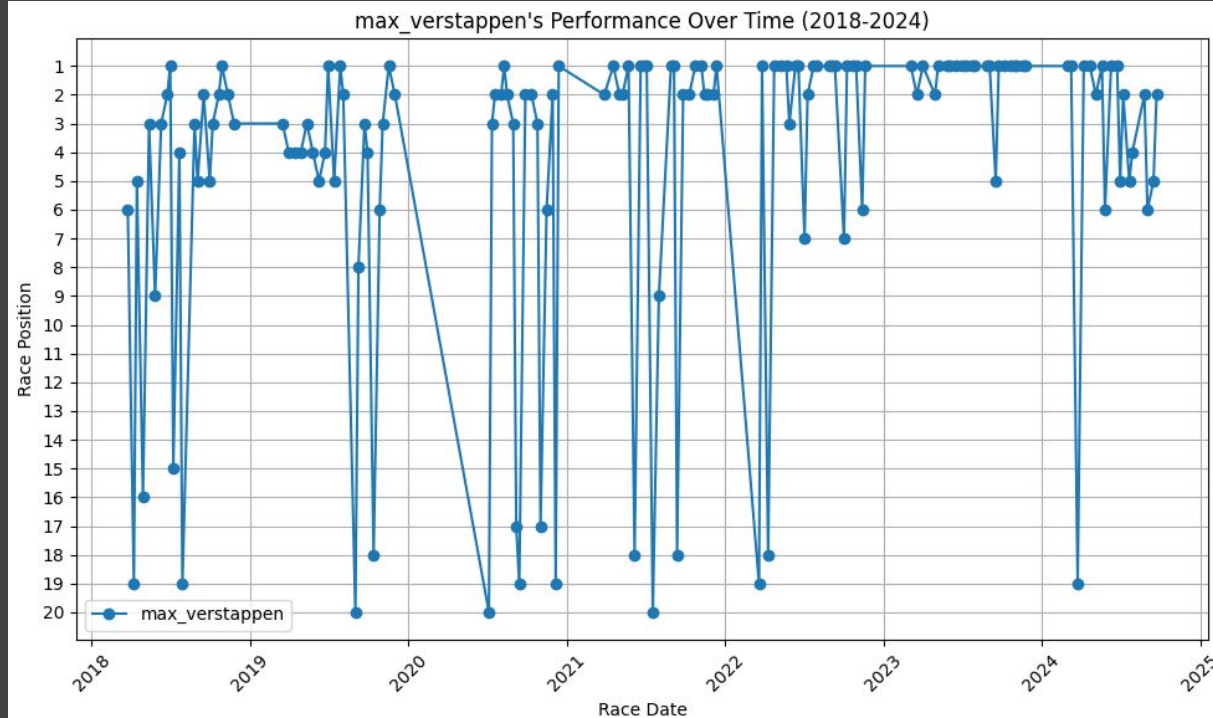
Attribute Analysis

Attribute Name	Range of Valid Values	Mean	Median	Mode	Unit
Air Temperature	9.45-36.57	23.59	23.20	9.5	Celsius
Relative Humidity	7.11-96.44	53.33	54.53	7.11	%
Air Pressure	779.52-1023.13	988.04	1006.33	779.52	mbar
Rainfall	True/False	-	-	-	boolean
Track Temperature	15.21-54.53	35.79	35.75	15.21	Celsius
Wind Speed	0-359	1.66	1.52	0.27	Degrees
Driver ID	1-99	27.5	20	10	-
Race Points	0.0-26.0	5.07	0.5	0.0	-
Grid Position	0.0-20.0	10.07	10.0	1.0	-

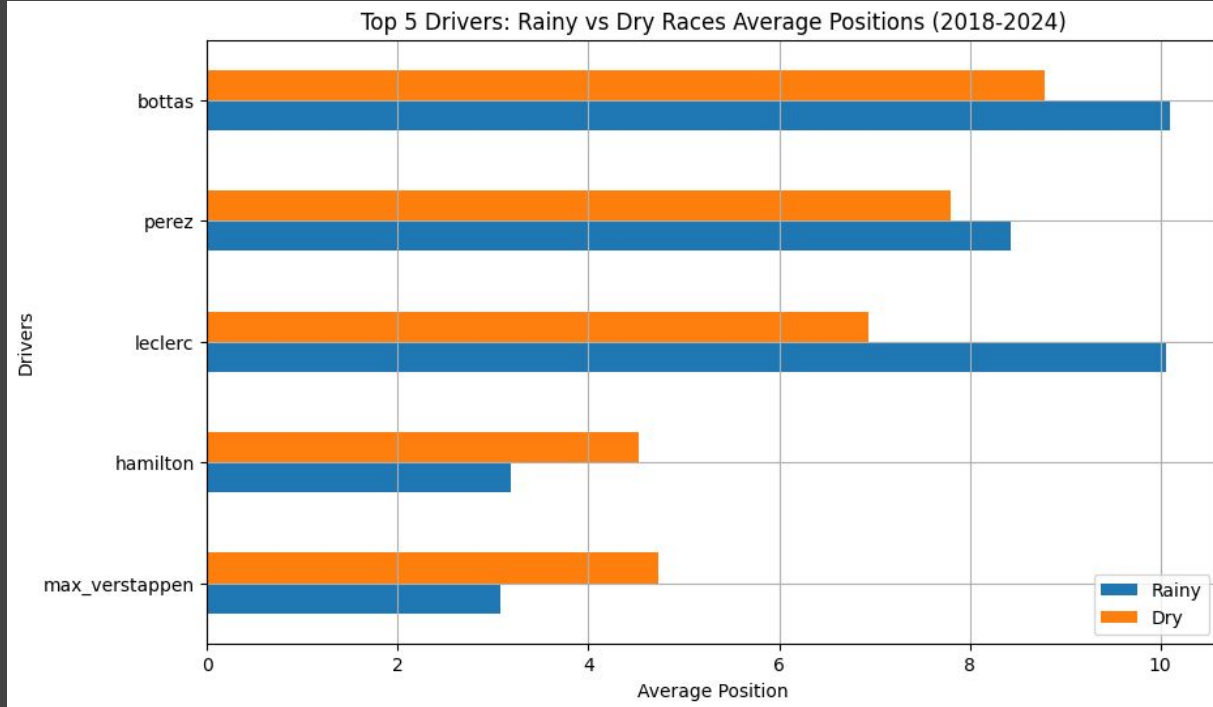
Visualization 1: Top 5 drivers sorted by total points earned (2018-2024)



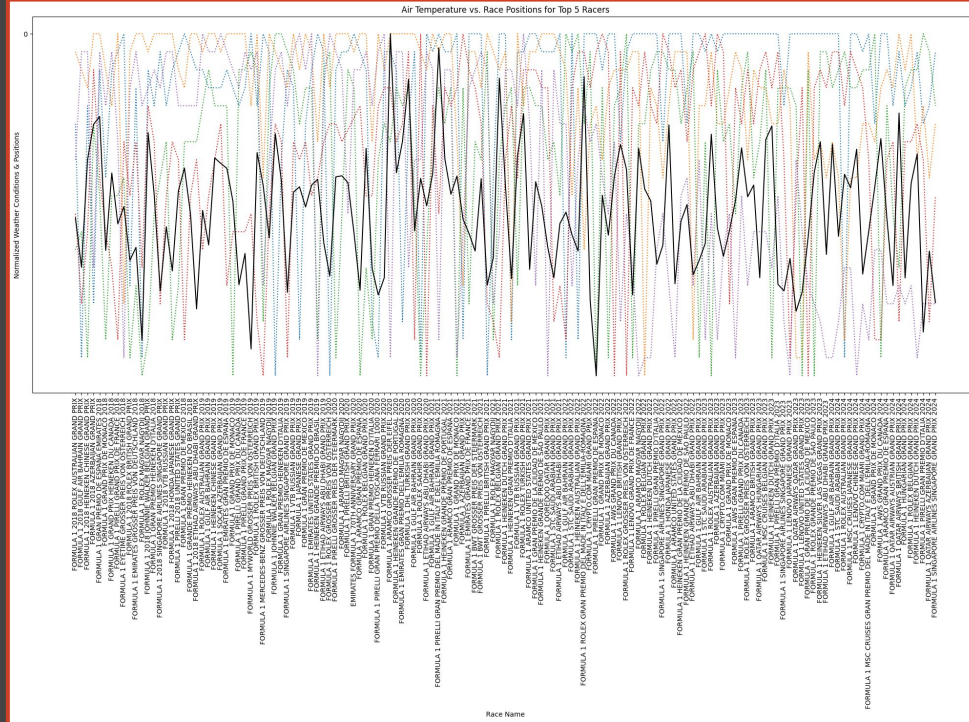
Visualization 2: Top 5 drivers' performance over time (2018-2024)



Visualization 3: Rainy vs Dry Races Average Position (2018-2024)



Visualization 4: Individual Weather Conditions vs. Top 5 Drivers' Positions

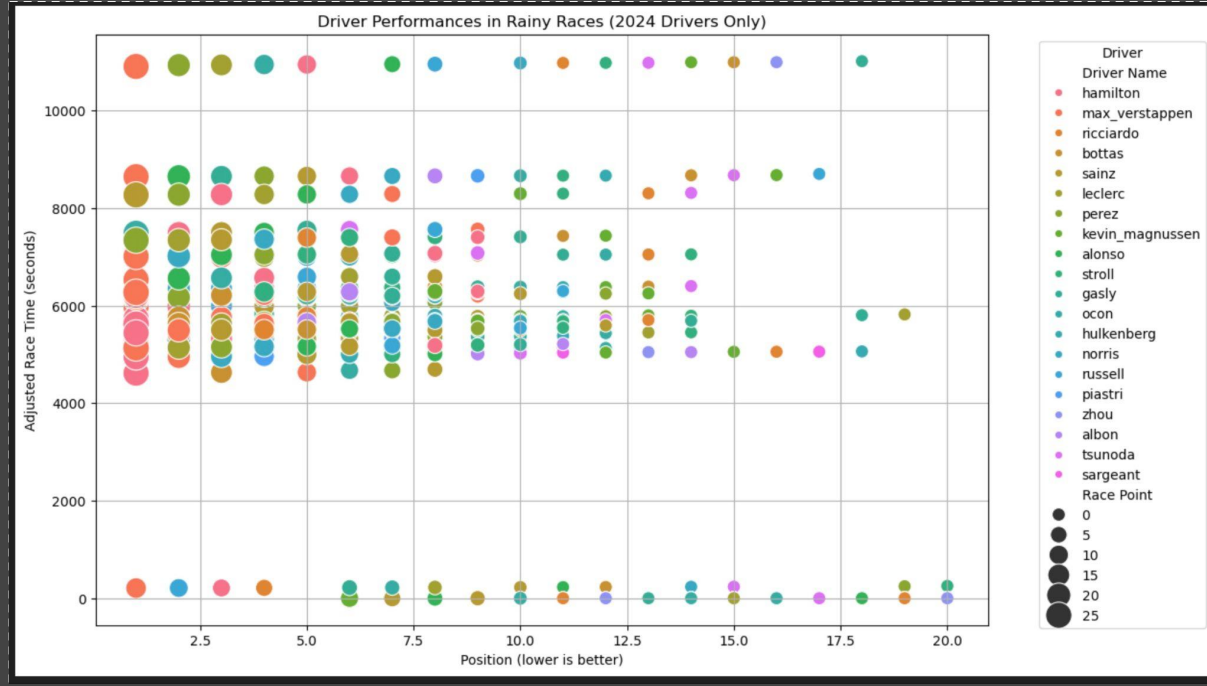


Note : the positions are displayed against each weather condition in our dataset. There are four additional drivers included in the analysis (the code loops through all five drivers)

What did we notice? (based on visualizations)

- **Sorting out top 5 drivers**
 - Over the years, while **Verstappen** constantly stays high positions during each race, other drivers like **Leclerc** or **Perez** showed more variability in terms of their performance (some high, some low positions).
 - As mentioned above, **Verstappen** started with high points but not always at first place. However, since 2021, he constantly remained in first place or close to it in most races.
 - **Perez's** positions showed more variability in his performances (2018 to early 2020). However, since 2020, while his performance remained inconsistent, there were clear signs of improvements over time.
- **Driver's performance over the time**
 - **Leclerc's** performance shows an initial struggle in earlier seasons, but there is a noticeable improvement over the years. On the other hands, **Bottas'** performance declined from 2022, represented by a clear downward trend in his results.
- **Rainy vs Dry Races Average Position**
 - **Verstappen** and **Hamilton**, both their performances were slightly better in rainy conditions, but still maintained consistent high positions. Both still perform well in both types of races.
 - **Perez's'** performance has a very minor gap between dry and rainy races, demonstrating his ability to maintain his performance in both scenarios.
- **Individual Weather Conditions vs. Top 5 Drivers' Positions**
 - The race positions tend to fluctuate a little more widely in races with lower track temperatures. This might indicate that colder conditions lead to more unpredictable race outcomes.
 - Most driver seem to perform better with higher wind speeds.

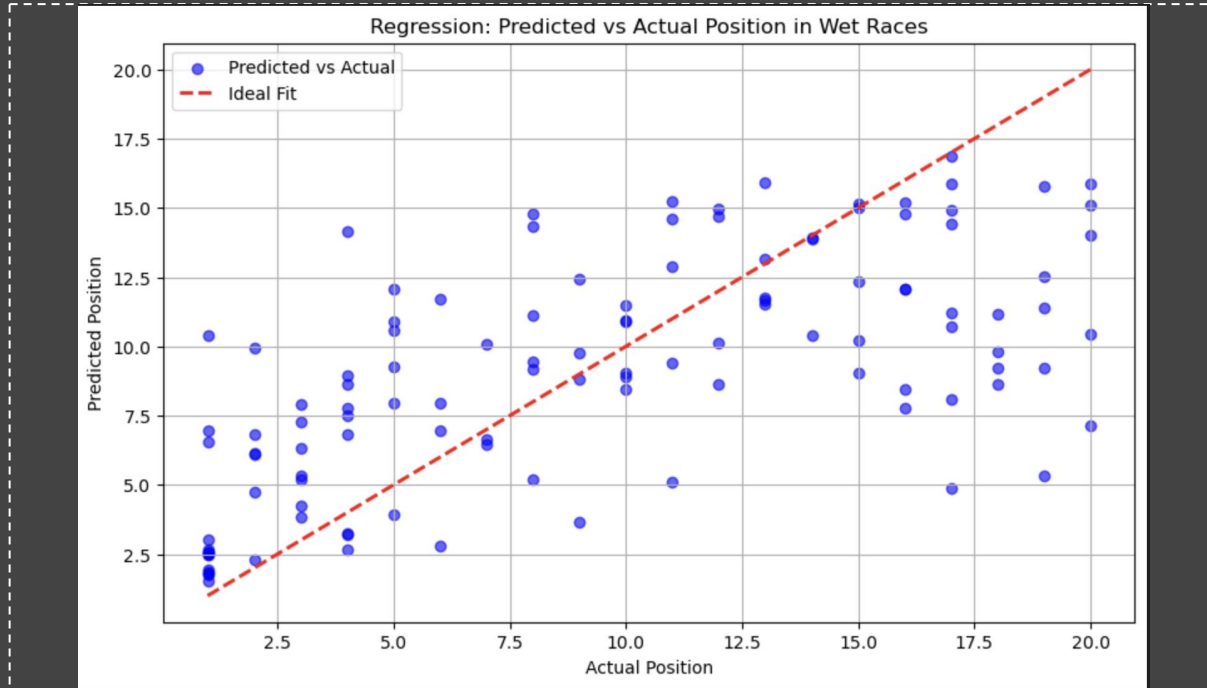
Data Mining Suitability - Frequent Patterns in Wet Races



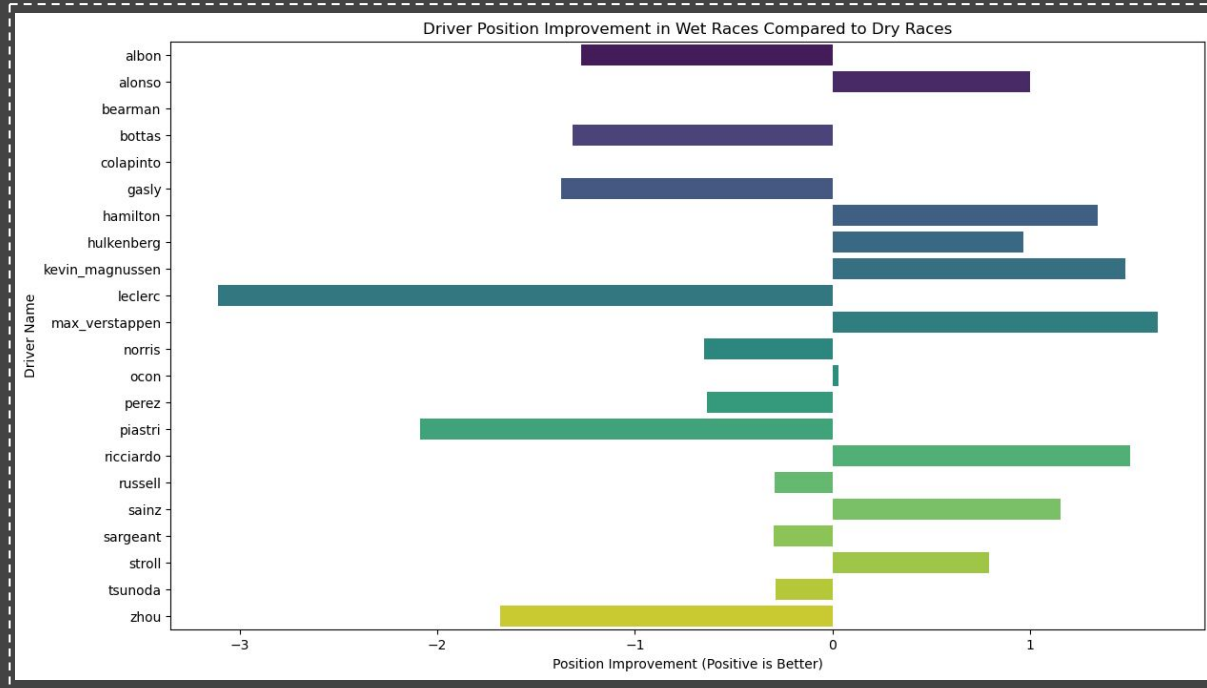
Data Mining Suitability - Clustering Performance of Drivers Across Race Finish Times



Data Mining Suitability - Regression on Performance in Wet Races



Data Mining Suitability - Visualization on Performance in Wet vs. Dry Races



Data Mining Suitability - Takeaways

- **Regression**

- Regression Model metrics show poor performance, so predictions made may be off, need better features to train on
- Group of drivers who win in wet races that would normally finish lower
- Group of drivers who finish lower or DNF that would normally finish higher

- **Clustering**

- Distinct set of drivers who win consistently or are high finishers
- Group of drivers who DNF, in which the drivers in this cluster could be anyone
- Group of low position finishers no matter the length of the race

- **Frequent Patterns**

- Hamilton and Verstappen are the best wet weather drivers, which correlates with overall performance
- Some drivers excel in wet weather, some drivers do poorer

Thank You

